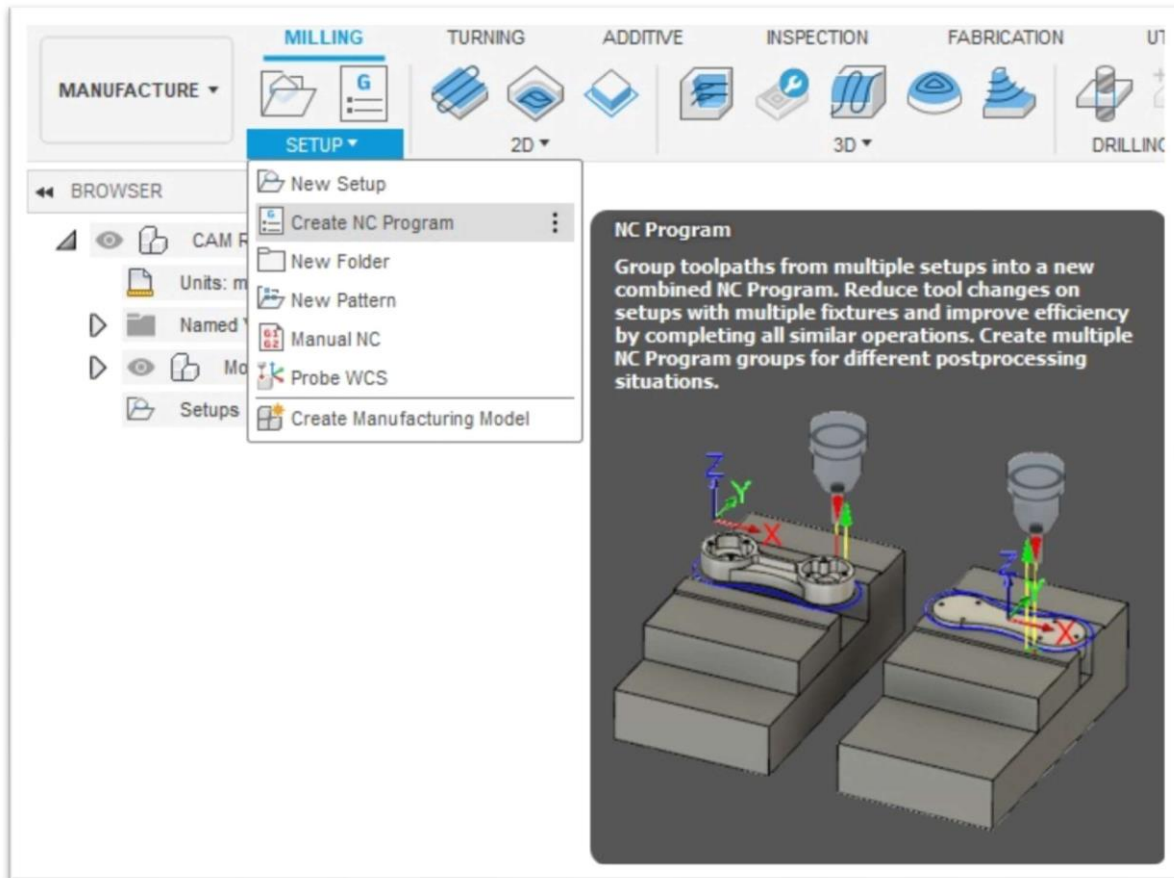


KUKA

ROBOT POSTPROCESSOR MANUAL

1. Create NC Program

NC Program can organize several Setups into a single NC program output, writing multiple toolpaths when needed:



It's important to use NC Program for writing your nc codes.

The NC Program dialog has an ordered tab to manage all the properties you need for the KUKA nc code.

2. NC Program Settings properties

This is an overview of all the properties available for KUKA output:

Post properties

▼ Process

End-effector state

OFF

Robot head angle

30

▼ Configuration

First point move type

Joints

Robot joint 1

5

Robot joint 2

-80

Robot joint 3

100

Robot joint 4

75

Robot joint 5

-60

Robot joint 6

-65

Robot 'Status' configuration

B110

Robot 'Turn' configuration

B110010

▼ Parameters

Flip tool frame

☒

Robot acceleration/deceleration (%)

20

Robot look-ahead (ADVANCE)

3

Robot PTP velocity (%)

25

Robot path smoothing (APO.CDIS) (mm)

2

▶ ☐ FFF Settings

▶ ☐ General

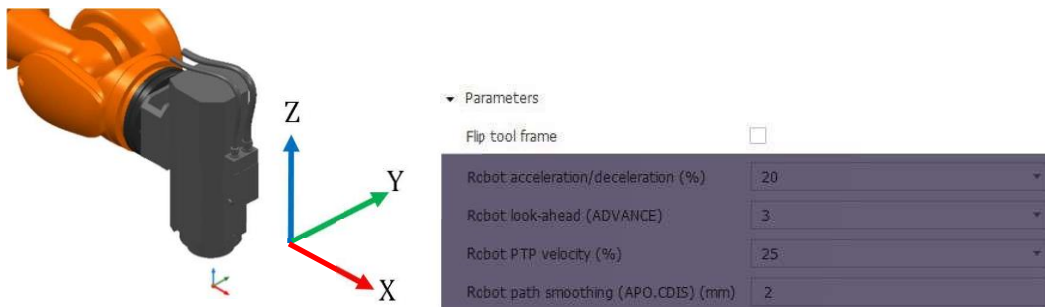
▶ ☐ Built-in

3- Confirm the tool orientation on the robot

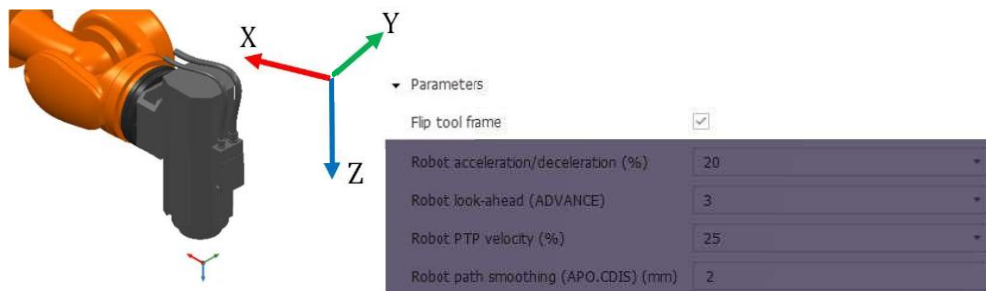
The robot can be driven manually along the tool coordinate system, this is one option to check the orientation of the tool workplane.

- Select the tool coordinate system
- Select the appropriate tool number to jog
- Use the teach pendant/enabling device to drive the robot along each axis individually
- This is a good way to check the orientation of X, Y and Z axis of the tool workplane.

If Z+ is pointing up along the tool axis and the X+ is pointing in front, **use No Flip** option:



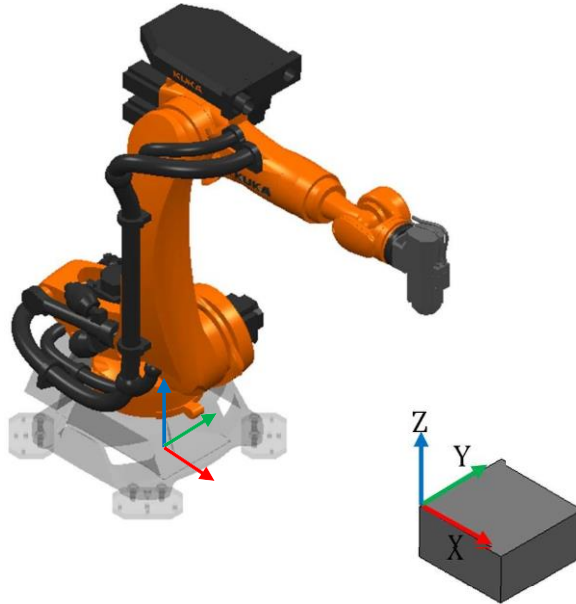
If Z+ is pointing down along the tool axis and X+ is pointing backwards, **use Flip** option:



Note: if the tool orientation is not one of the above, Autodesk CAM post will not support the application correctly.

4 - WCS setup (workplane)

On the KUKA robot it is possible to define a coordinate system on the part which is known as a **Base Data (Base)**. The **Base Data**, located on the part/block, will be referenced from the zero of the robot, which is located at its base.

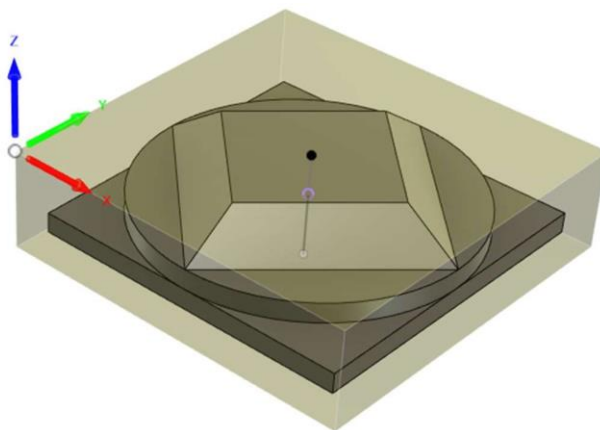
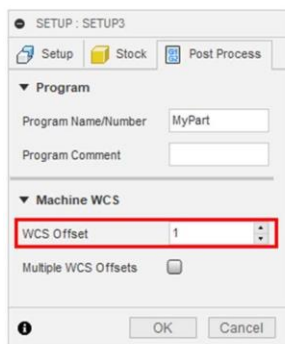


When a **Base** is defined on the part, the robot will have defined a:

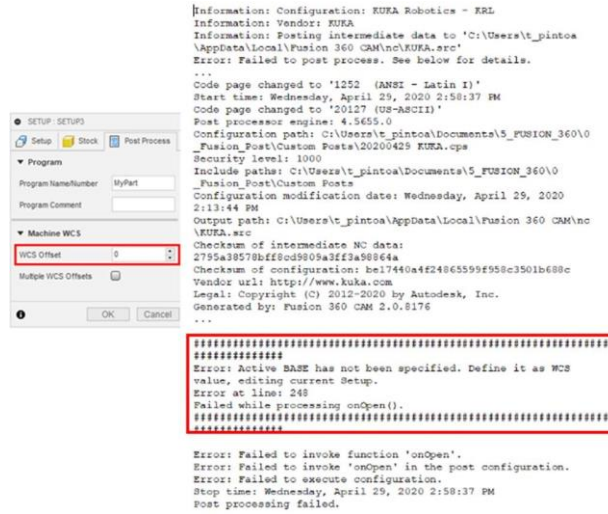
- Number
- Position & Orientation

To run a toolpath successfully on the robot, users must ensure the robot Base Data and the WCS in Autodesk CAM are in the same location and orientation.

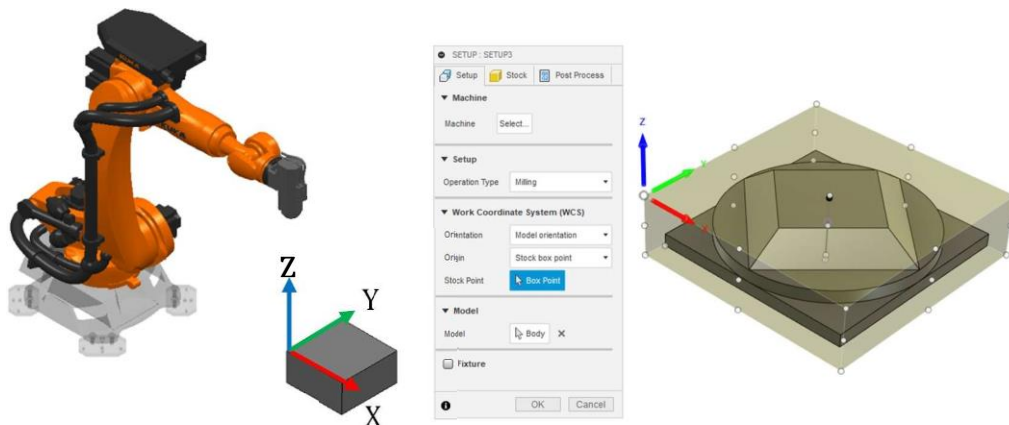
The WCS Offset number in Autodesk CAM defines the **Base** number, select the number defined on the robot.



Note: WCS 0 cannot be used, if zero is selected an error will be raised while post processing and no output will be written.



Use the WCS setup menu to replicate the location and orientation of the **Base Data** on the part.

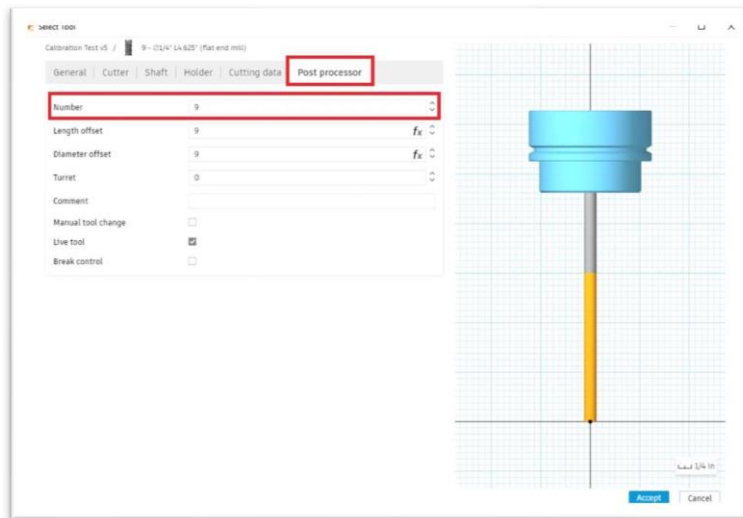


The robot can be jogged along the **Base Data** Coordinate System, this is one option to check the orientation of the **Base Data**.

- Select the **Base Data**
- Select the appropriate **Base Data** to jog
- Use the teach pendant/enabling device to drive the robot along each axis individually
- This is a good way to verify the orientation of X, Y and Z axis of the **Base Data**
- You can also jog the robot to the location of the origin and visualize the coordinates of the **Base Data**

5 - Define Tool Number (replicate settings on the robot)

The tool number is defined via the Tool Post Processor menu.

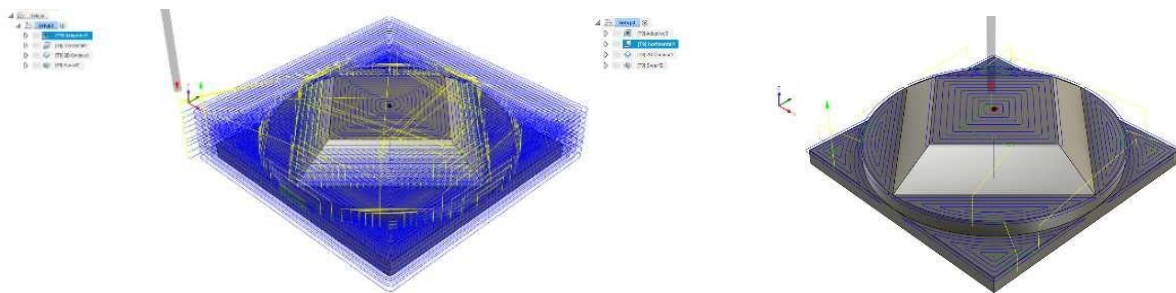


The tool will be declared in the main file, before the toolpath file is called.

```
; Set Tool
BAS (#TOOL, 9)
$TOOL = TOOL_DATA[9]
```

6 - Create a toolpath

Follow the usual steps inside Autodesk CAM in order to generate one or more toolpaths in your setup.



7 - Post processing

Autodesk CAM post-processor generates a few files.

- A pair of files is created, this means for each file a **.src** and a **.dat** file are generated.
- **Toolpath** files are named using the toolpath name given in Autodesk CAM adding the prefix 's'. File name should **NOT** start with a number and should not contain any special characters.
- A main file is generated, this will manage the call of the toolpath(s) and define tool and base used. The main file uses the name defined in the post processor menu or in the setup.



Once ready to post process, some post-processor properties need to be defined before output files can be generated:

Process properties

- End-effector state: This option is for all the non-subtractive operations when the end-effector needs to be turned on/off.



Codes to use end-effectors are non-standard and they must be customized according to the robot integration. The postprocessor simply writes these 2 lines as a comment:

==> END EFFECTOR ON: DEFINE YOUR CODE HERE IN THE POST

and

==> END EFFECTOR OFF: DEFINE YOUR CODE HERE IN THE POST

You can define your proper instructions inside the postprocessor editing these lines.

- Head angle: This allows the user to enter an angle of rotation around the tool axis, this will effectively rotate the spindle, the angle will be kept throughout the entire toolpath. This angle is relative to the X axis of the WCS defined on the part.

▼ Process

End-effector state

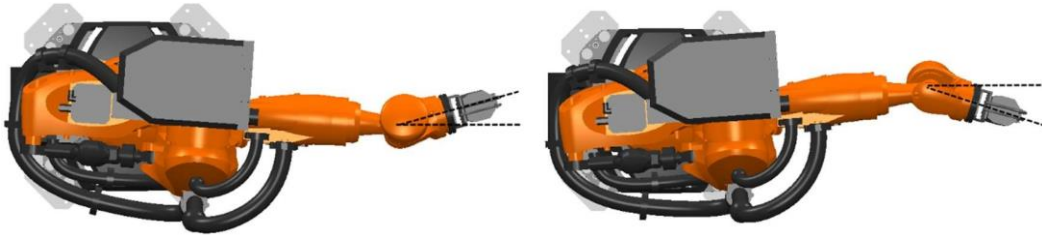
OFF

▼

Head angle

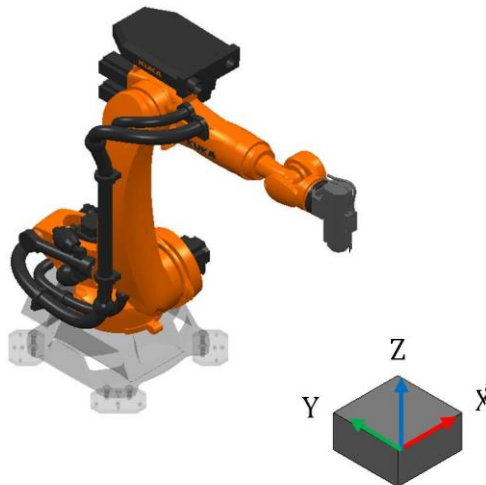
30

Below an example of a 15 degrees (left) and -15 degree (right) relative to the X axis, for a tool defined with Z+ going up the spindle.

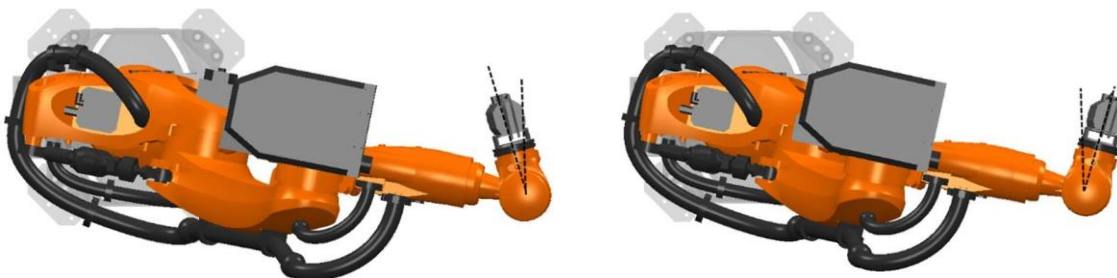


If the X axis is defined on another orientation the spindle angle will be relative to this orientation.

Note: X axis on the **Base Data** is NOT pointing forward in this example



15 degree (left) and -15 degree (right)



Configuration properties

- **First point move type:** Allows the user to choose the first move type using robot joints or a cartesian position.

Configuration

First point move type

↓

Cartesian

Joints

Robot joint 1

5

Robot joint 2

-80

Robot joint 3

100

Robot joint 4

75

Robot joint 5

-60

Robot joint 6

-65

Robot 'Status' configuration

B110

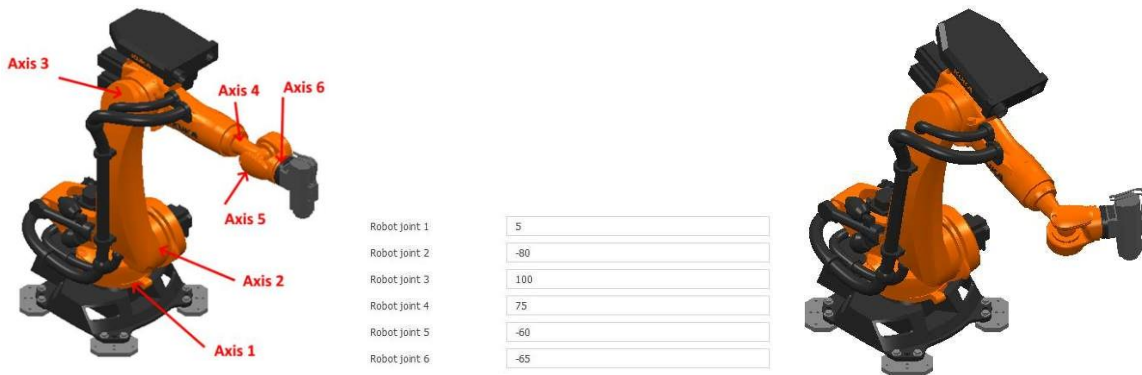
Robot 'Turn' configuration

B110010

Default is joint move, below the axis position defined by a **PTP** move where A1=5, A2=-80, A3=100, A4=75, A5=-60, A6=-65 is the axis position.

```
PTP {A1 5,A2 -80,A3 100,A4 75,A5 -60,A6 -65,E1 0,E2 0,E3 0,E4 0,E5 0,E6 0}
```

- **Robot joint:** The joint values are defined in the form below.



If cartesian position is defined, the robot will move to the first toolpath point using a **PTP (with Cartesian coordinates)** relative to the WCS/**Base Data** defined on the part, this will require reading the robot configuration from the teach pendant and updating fields *Status* and *Turn* below.

```
PTP {X 8.432,Y -62.821,Z 15.000,A -150.000,B 0.000,C 180.000,E1 0,E2 0,E3 0,E4 0,E5 0,E6 0,S'B110',T'B110010'}
```

Configuration

First point move type	Joints
Robot joint 1	5
Robot joint 2	-80
Robot joint 3	100
Robot joint 4	75
Robot joint 5	-60
Robot joint 6	-65
Robot 'Status' configuration	B110
Robot 'Turn' configuration	B110010

- Robot 'Status' configuration: This allows the user to define the Status configuration.
- Robot 'Turn' configuration: This allows the user to define the Turn configuration.

Both properties should be used when using cartesian position as the first point move type since the robot configuration must be defined in order to have a unique solution.

PTP {X 8.432,Y -62.821,Z 15.000,A -150.000,B 0.000,C 180.000,E1 0,E2 0,E3 0,E4 0,E5 0,E6 0 S'B110',T'B110010'}

An example of the same tool tip position & orientation but with 2 different robot configurations. Notice that robot axis 4, 5 and 6 are reversed.



Status=2 ('B010') Turn=10 ('B001010')



Status=6 ('B110') Turn=50 ('B110010')

Parameter properties

Parameters	
Flip tool frame	☒
Robot acceleration/deceleration (%)	20
Robot look-ahead (ADVANCE)	3
Robot PTP velocity (%)	25
Robot path smoothing (APO.CDIS) (mm)	2

- Robot acceleration/deceleration (%): This will set the Robot acceleration/deceleration.

Parameters	
Flip tool frame	☒
Robot acceleration/deceleration (%)	20
Robot look-ahead (ADVANCE)	3
Robot PTP velocity (%)	25
Robot path smoothing (APO.CDIS) (mm)	2

- Robot look-ahead: This allows the end user to define how many points the controller reads ahead.

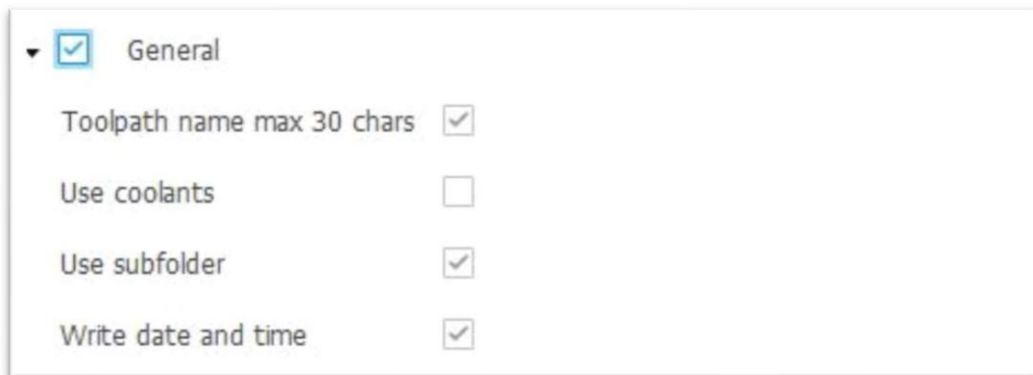
Parameters	
Flip tool frame	☒
Robot acceleration/deceleration (%)	20
Robot look-ahead (ADVANCE)	3
Robot PTP velocity (%)	25
Robot path smoothing (APO.CDIS) (mm)	2

- Robot PTP velocity: This defines the robot velocity on PTP move (first move).



- Robot path smoothing: This defines the robot smoothing distance (see KUKA technical documentation for details)

General properties



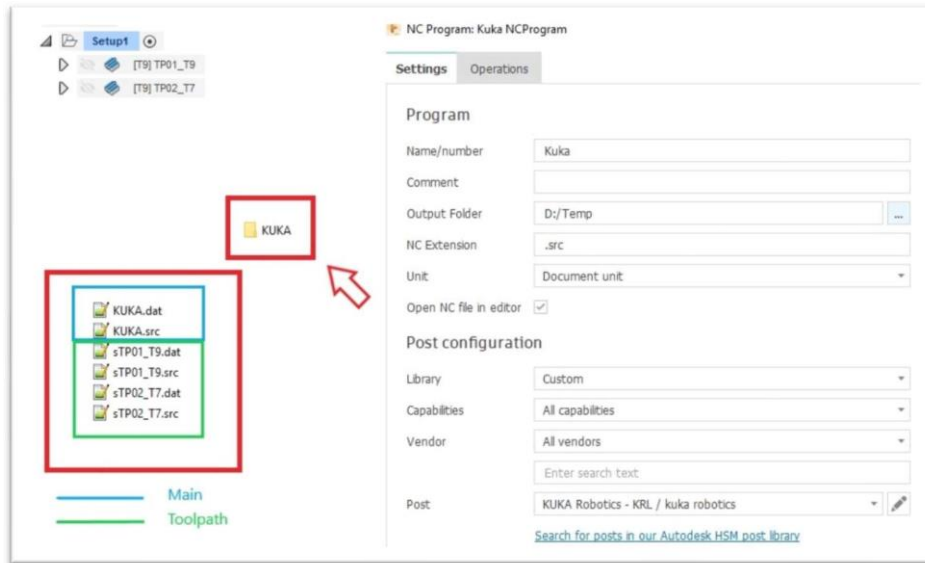
- Use coolants: When this option set as No (default) post-processor will ignore Coolant definition set in Autodesk CAM.
- Use subfolder: Default is to write out all data in a sub folder.

A dummy file with standard information is created after posting if you chose to Use subfolder. It contains the name of the directory where you can find your nc files. It is called as the Program Name in NC Program form. This is a dummy file example:

```
This is a dummy file.
Your program files are located here: C:\Users\xyz\AppData\Local\Fusion 360 CAM\nc\1001
```

- Toolpath name max 30 chars: Default is to check each toolpath name length. An error will be raised when length is more than 30 char.
- Write date and time: Writes date & time of file creation in the main .MOD file:
; Creation date: Wednesday, April 29, 2020 15:40:10

Below an example setup with two toolpaths, KUKA is the name of the folder and the Main file. The toolpath files use the toolpath names with the prefix 's'.



6 – General information

For more information get help or post your questions on the forum:

<https://forums.autodesk.com> Select "Fusion 360" and then "Fusion 360 Manufacturing"

In KUKA Robot Language (KRL files), the default unit of speed is millimeters per second: **m/s**.
\$VEL.CP=0.10; means robot TCP speed = 1.6667 mm/s (≈100 mm/min)

If the Autodesk CAM session is running in inches, the output file will still be written in mm as this is the unit system used on robots.

IMPORTANT: Please remember that program and toolpath names should not contain any symbol or special character.

7 – Program sample

Autodesk CAM post-processor generates a few files, as described above on point 5.

This is an example of KUKA postprocessor output. The example has one only toolpath.

```
--- main.src file Start ---
&ACCESS RVO
DEF KUKA()
;FOLD INI
    ;FOLD BASISTECH INI
        GLOBAL INTERRUPT DECL 3 WHEN $STOPMESS==TRUE DO IR_STOPM ( )
        INTERRUPT ON 3
        BAS (#INITMOV,0 )
    ;ENDFOLD (BASISTECH INI)
    ;FOLD USER INI
        ;Make your modifications here
    ; Robot Postprocessor

    ;ENDFOLD (USER INI)
;ENDFOLD (INI)

; Generated by AUTODESK Fusion 360 CAM 2.0.8176
; Creation date: Wednesday, May 06, 2020 18:11:22

; Set PTP velocity and acceleration
$VEL_AXIS[1] = 25
$VEL_AXIS[2] = 25
$VEL_AXIS[3] = 25
$VEL_AXIS[4] = 25
$VEL_AXIS[5] = 25
$VEL_AXIS[6] = 25

; To set 'Tool On Robot' Mode
$IPO_MODE = #BASE

; Set smoothing value
$APO.CDIS = 2

; Set $ADVANCE value (number of points read in advance)
$ADVANCE = 3

; Set BASE
$BASE = BASE_DATA[54]

; Load Tool
;TOOL_CHANGE(1)

; Set Tool
BAS(#TOOL,1)
$TOOL = TOOL_DATA[1]

; Spindle ON/speed
;SPINDLE_ON(4850)

sTP01()

; Spindle OFF
;SPINDLE_OFF()

END
--- main.src file End ---
```

--- main.dat file Start ---

```
&ACCESS RVO
DEFDAT KUKA
;FOLD EXTERNAL DECLARATIONS;{%PE}%MKUKATPBASIS,%CEXT,%VCOMMON,%P
;FOLD BASISTECH EXT;{%PE}%MKUKATPBASIS,%CEXT,%VEXT,%P
EXT BAS (BAS_COMMAND :IN,REAL :IN )
;ENDFOLD (BASISTECH EXT)

EXT sTP01()

;ENDFOLD (EXTERNAL DECLERATIONS)
ENDDAT
--- main.dat file End ---
```

--- stoolpath.src file Start ---

```
&ACCESS RVO
DEF sTP01()
; Generated by AUTODESK Fusion 360 CAM 2.0.8176
; Creation date: Wednesday, May 06, 2020 18:11:22
;FOLD sTP01
; Start position (joint) = {A1 5,A2 -80,A2 -80,A3 100,A4 75,A5 -60,A6 -65}
; Tool Number = 1
; Spindle Speed = 4850 RPM
; Program file name = sTP01.src

PTP {A1 5,A2 -80,A3 100,A4 75,A5 -60,A6 -65,E1 0,E2 0,E3 0,E4 0,E5 0,E6 0}
$VEL.CP=0.02
LIN {X 64.000,Y -32.892,Z 15.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X 64.000,Y -32.892,Z 5.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X 64.000,Y -32.892,Z 1.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X 63.732,Y -32.892,Z 0.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X 63.000,Y -32.892,Z -0.732,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X 62.000,Y -32.892,Z -1.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X 51.000,Y -32.892,Z -1.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X -51.000,Y -32.892,Z -1.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X -53.384,Y -32.515,Z -1.000,A -150.000,B 0.000,C 180.000} C_DIS
LIN {X -55.534,Y -31.419,Z -1.000,A -150.000,B 0.000,C 180.000} C_DIS
;ENDFOLD
--- stoolpath.src file End ---
```

--- stoolpath.dat file Start ---

```
&ACCESS RVO
DEFDAT sTP01
;FOLD EXTERNAL DECLARATIONS;{%PE}%MKUKATPBASIS,%CEXT,%VCOMMON,%P
;FOLD BASISTECH EXT;{%PE}%MKUKATPBASIS,%CEXT,%VEXT,%P
EXT BAS (BAS_COMMAND :IN,REAL :IN )
;ENDFOLD (BASISTECH EXT)

;ENDFOLD (EXTERNAL DECLERATIONS)
ENDDAT
--- stoolpath.dat file End ---
```